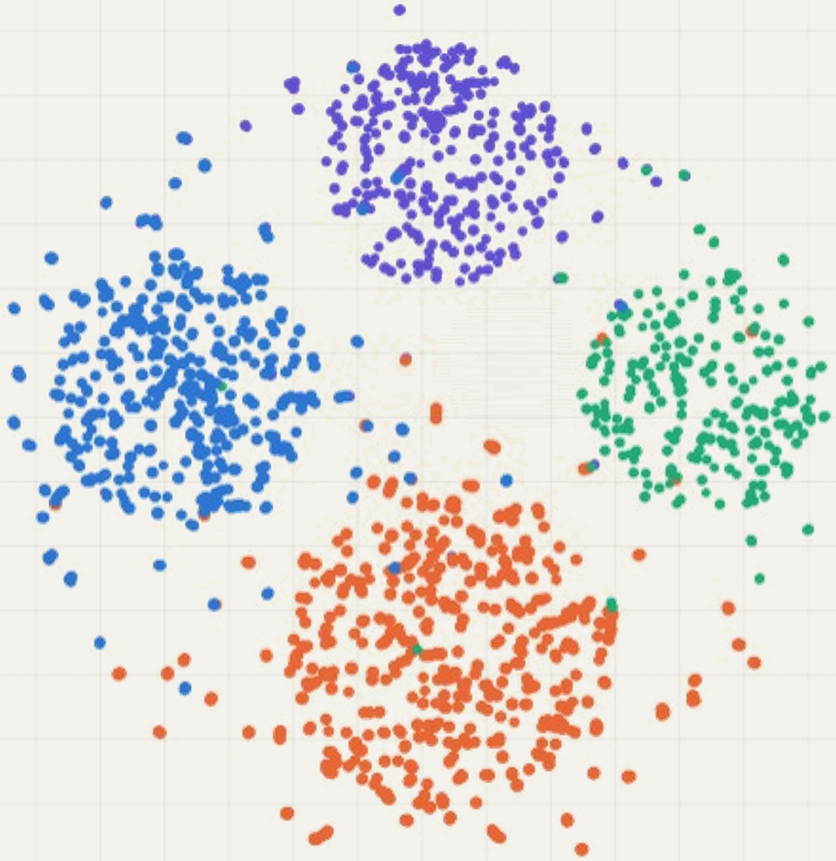
TEAM DUMBOCRACY

Clout.

An Election Game

NAME	ROLL NUMBER
Pakki Sai Kaushik	AI23BTECH11017
G Madhava Asrith Murthy	AI23BTECH11007
Vuppula Hemanth Reddy	AI23BTECH11033
Vishank Singh	AI23BTECH11032
B U Deekshith Patel	AI23BTECH11003



MOTIVATION

Campaigns spend big to move opinions they can't see

1 **Opinions drift** DEGROOT CONSENSUS

People slowly come to think like their friends.

2 **Influence spreads** INDEPENDENT CASCADE

A message passed on can die out or take off.

3 **Campaigns work blind** EPISTEMIC LOGIC

A party never knows exactly what voters want.

INDIA, 2019 ELECTION

₹60,000 cr

spent on one general election, mostly to change what people think.

Centre for Media Studies

THE GAME

Win a majority in five rounds, without seeing the voters

01

Setup

A hidden electorate of 10,000 voters is drawn. You get 500 coins.

02

Survey

Buy questions, then commit a **manifesto**: fixed for the game.

03 · ×5 rounds

Campaign

Hire party workers, air media. Voters drift towards their friends.

repeat each round

04

Election

More than half the votes wins. Then the hidden truth is revealed.

10 coins
Survey question

20 coins
Party worker

~50 coins
Media airing

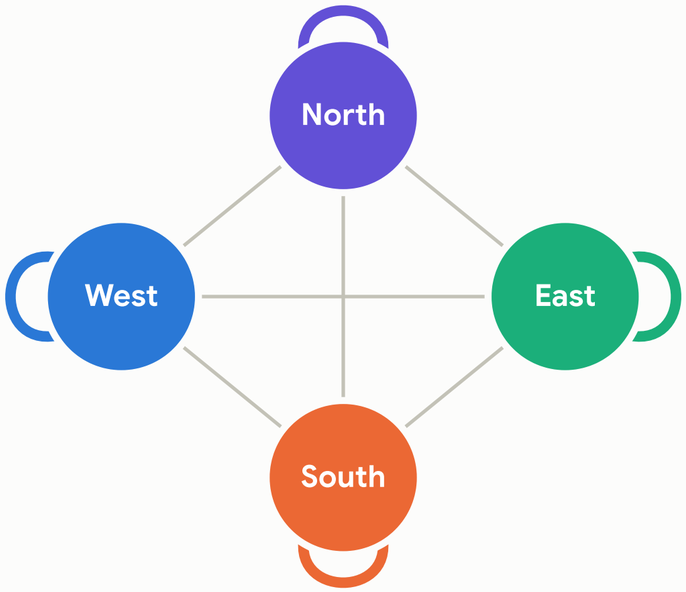
MANIFESTO = A BUDGET SPLIT



Education · Welfare · Agriculture · Industry · Environment

THE ELECTORATE

Four regions, one friendship network, hidden tastes



Thick loops: ~¾ of friendships stay inside a region.

WANTS

A budget split

Each region has a hidden archetype, e.g. *agrarian* or *industrial*.

MOVED BY

Friends, media or canvassing

Each region has a hidden personality.

ZEALOTS

5% never change

But their friends still listen to them.

VOTES

Only when close enough

Leans your way once it agrees with **82.5%** of your budget.

THE ELECTORATE · IN MATHS

How we build a world

REGIONS	$n_r = m + \big[\mathrm{Mult}(N - 4m,\ \pi)\big]_r, \quad m = \lceil 0.15N \rceil$	Each region gets **≥ 15%**; the rest split at random
FRIENDSHIPS	$P_{rr} = s\,p_r, \ \ P_{rs} = s\,p_{rs}, \quad p_r \sim U[0.5, 0.9],\ p_{rs} \sim U[0.01, 0.2]$	Stochastic block model; **s** sets mean degree 30
WANTS	$v_i(0) \sim \mathrm{Dir}\big(\kappa\,\mu_{\sigma(r)}\big), \quad \kappa = 60, \ \ v_i \in \Delta^5$	Budget split around the region's hidden **archetype σ**
MOVED BY	$\alpha_i,\ \beta_i,\ \gamma_i \sim \mathrm{Beta}\big(\tau(r)\big), \quad \Pr[\text{zealot}] = 0.05$	Friends, media, canvassing via hidden **personality τ**
DRIFT	$v_i \leftarrow (1 - a_i - c_i)\,v_i + a_i\,\bar{v}_{\mathcal{N}(i)} + c_i\,b, \quad a_i = 0.15\,\alpha_i$	Damped **DeGroot** step; **c** is the campaign pull
VOTES	$P_i = \mathrm{sig}\big(20\,(S_i - 0.825)\big), \quad S_i = 1 - \tfrac{1}{2}\lVert v_i - b \rVert_1$	Agreement with manifesto **b** sets the vote chance

HIDDEN TYPE $\theta = (\sigma, \tau)$: $360 \times 24 = 8640$ worlds — What the player must infer from surveys and polls.

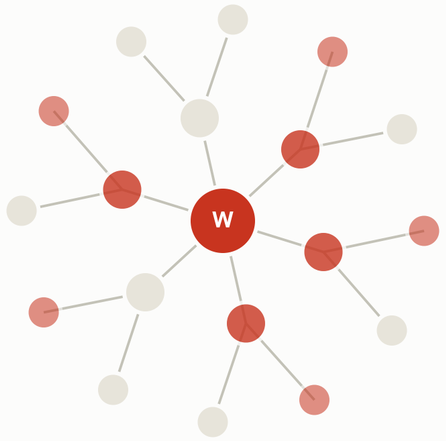
TWO WAYS TO SPEND

Workers start cascades. Media gives a nudge.

Workers `20 coins`

- A real voter, pinned to your manifesto
- Converts friends, **one hop per round**
- Hire early: more rounds to spread

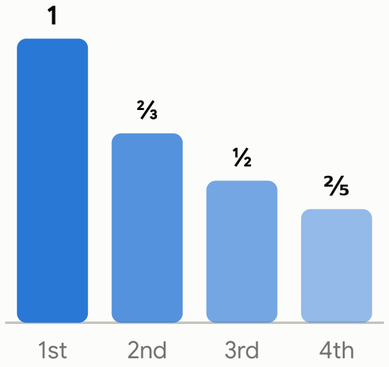
Takes off when $\lambda_{max}(A \cdot D_q) > 1$



Media `~50 coins`

- Reaches **60%** of a region at once
- A temporary pull, no network needed
- Each repeat airing is weaker

Strength ∝ `1 / (1 + 0.5·n)`



airing in the same region

LINK The manifesto matters for both: voters who already like it convert more easily.

EPISTEMIC & TEMPORAL LOGIC

Knowing a win exists ≠ knowing how to win

The player can't see the world, so we track every world that **matches what they have seen**.

K true in **every** such world: the player *knows* it

M true in **some** world: it is *possible*

After each game, the debrief shows these verdicts for every round, alongside the hidden truth.

Knows *that*

every world has some winning plan

	π1	π2	π3
w1	1	0	0
w2	0	1	0
w3	0	0	1

No single plan is safe

Knows *how*

one plan wins in every world

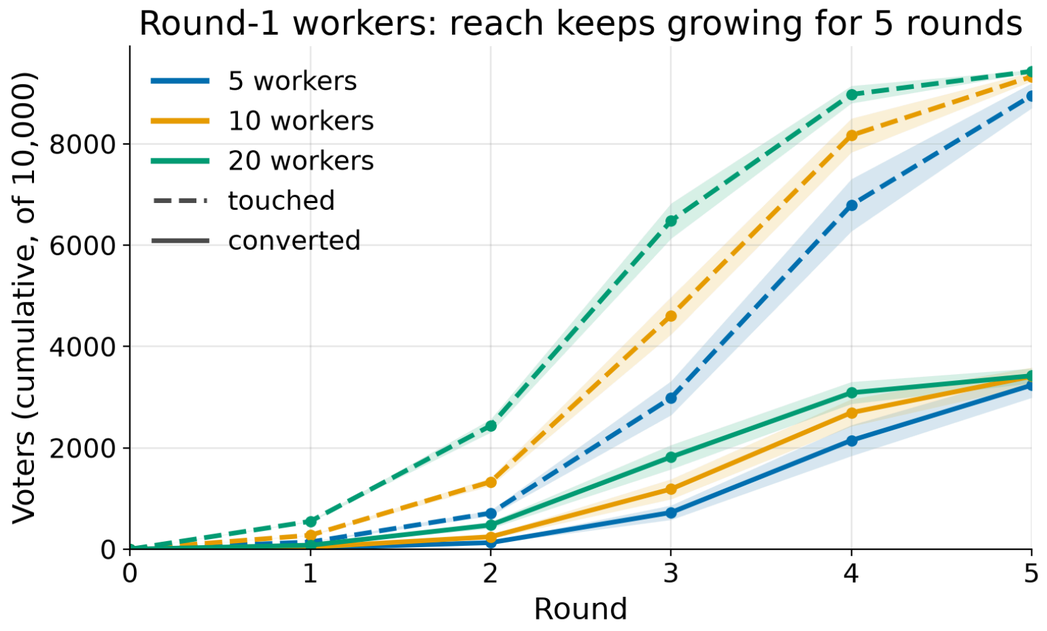
	π1	π2	π3
w1	1	1	0
w2	0	1	0
w3	0	1	1

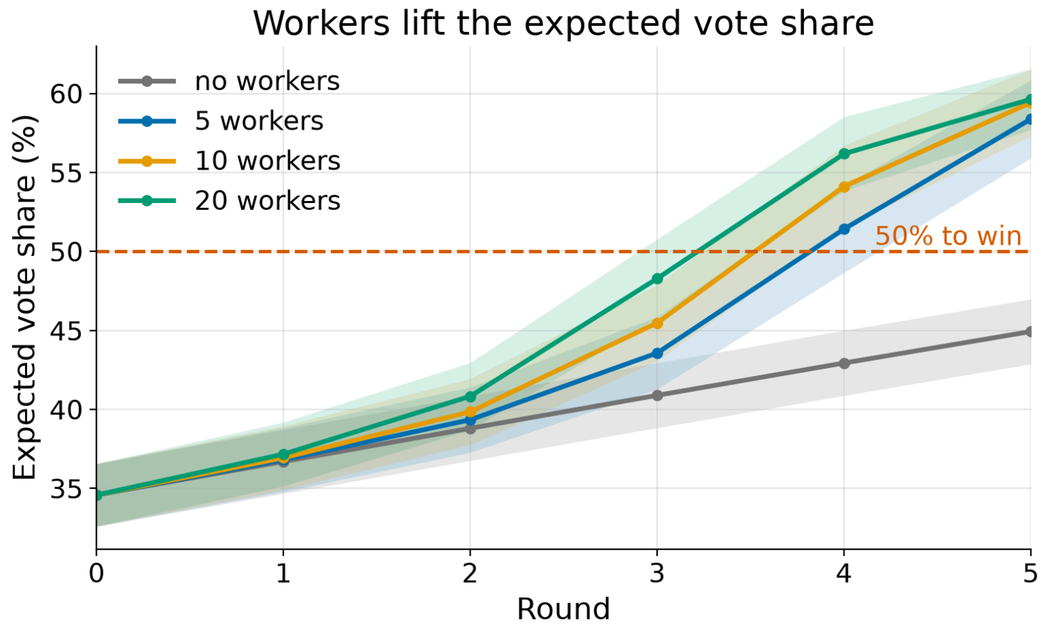
Play π2: guaranteed victory

Guaranteed ⇒ Knows a win exists ⇒ Possible neither arrow reverses

EXPERIMENT 1 · WORKERS BOUGHT IN ROUND 1 ONLY · 10,000 VOTERS · 50 WORLDS

A few early workers reach almost everyone





5 workers: 45% → 58% vote share

Win rate 28% → 82%. The cascade touches ~9,000 voters and converts ~3,200.

More workers: sooner, not bigger

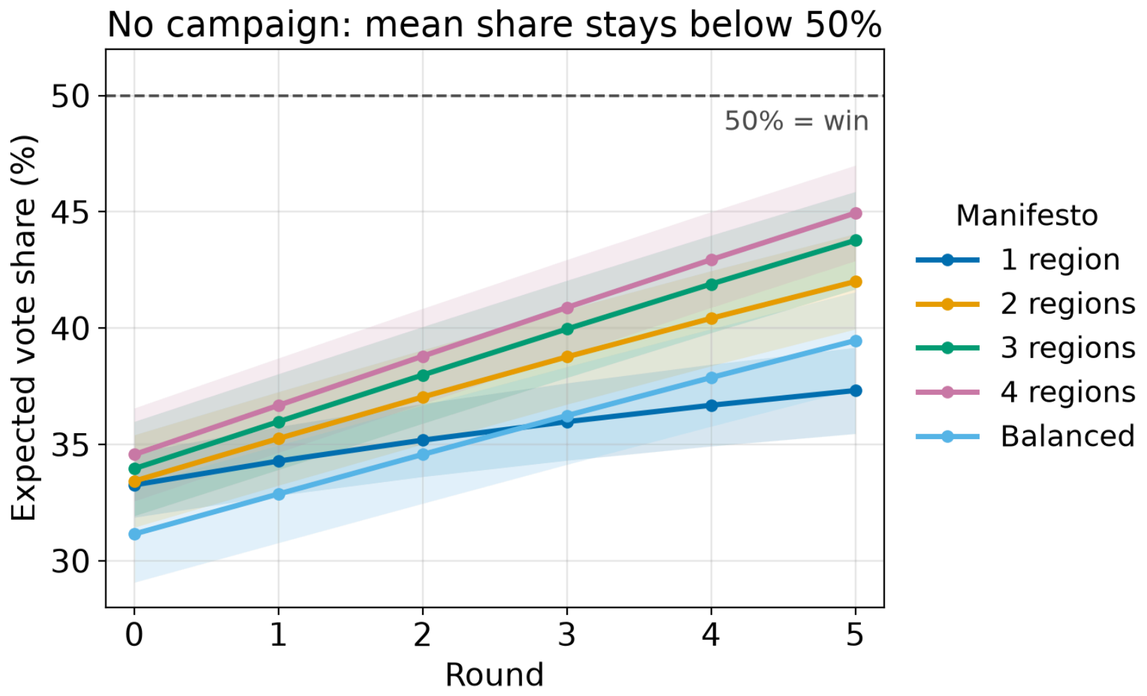
20 workers add only ~1 point over 5 by round 5; they mainly bring the gain forward.

The spread crosses regions

By round 5, about 30–37% of every region is converted.

EXPERIMENT 2 · FIXED MANIFESTOS, NO CAMPAIGN · 50 WORLDS

Blend more regions, win more votes

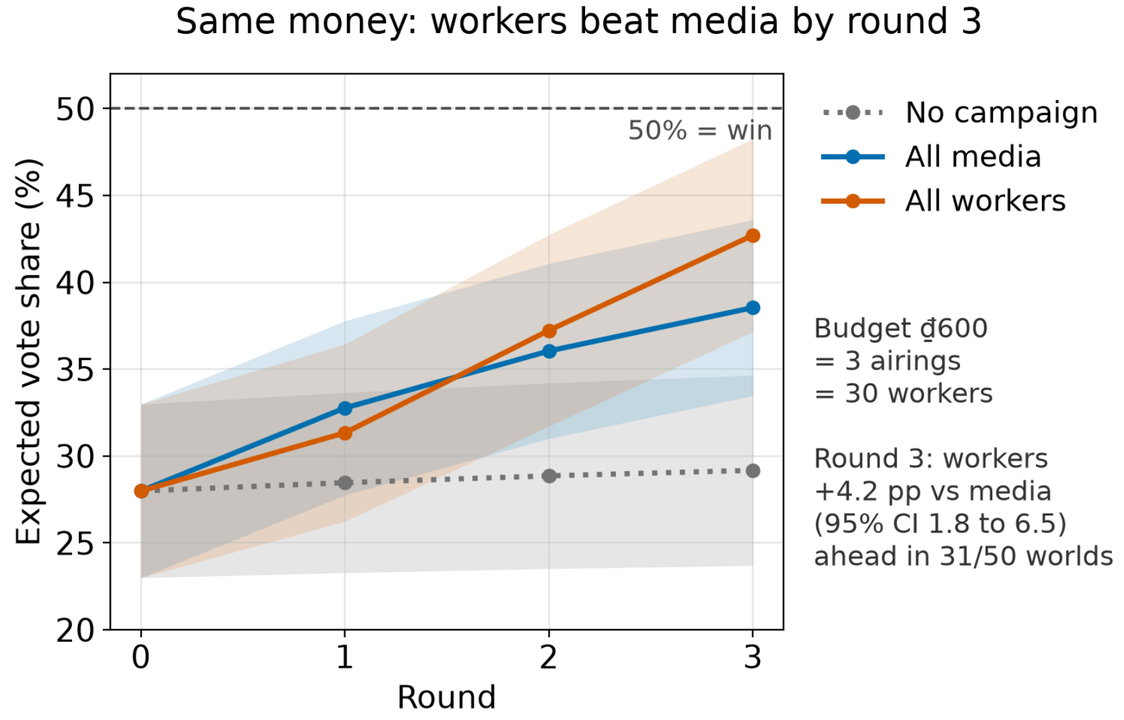


- **No manifesto wins alone.** Drift adds ~2 points a round, but none reaches 50% in the game.
- **The 4-region blend leads:** 44.9% at round 5 vs 37.3% for one region, and best in 37 of 50 worlds.
- **A flat 20% split starts worst** (31%) and stays below every blend.

2,500 voters · no workers, media or surveys · manifestos built from archetype blends, as in the game's "Blend my hunches".

EXPERIMENT 3 · ALL MEDIA VS ALL WORKERS · SAME ₫600 · 50 WORLDS

Same money: workers overtake media



- **Media leads after round 1,** but the worker cascade overtakes it from round 2.
- **Workers +4.2 points at round 3** (95% CI 1.8–6.5), ahead in 31 of 50 worlds.
- **Both beat doing nothing**: elections won 12 (workers), 6 (media), 5 (none).

2,500 voters in one region · 3 rounds · 20% manifesto · ₫600 = 3 airings = 30 round-1 workers. In the full game the best plans mix ~3 workers with early media.

OUR STRATEGY

Majority appeasement

LARGEST REGION

Appeased

Manifesto · light nudge

2ND LARGEST

Appeased

Manifesto · light nudge

3RD

Campaigned

Workers · Media

4TH

Campaigned

Workers · Media

1 **Win the majority with the manifesto**

Blend the archetypes of the two largest regions, weighted by size.

2 **Spend on the rest**

Workers and media go to the two smaller regions the manifesto doesn't reach.

3 **Keep the base**

A light nudge in the appeased regions stops drift from pulling them away.

LIMITATIONS

What our results do not tell you

Synthetic voters	Tuned so the game is playable, not fitted to a real electorate.
One party	No opponent, the manifesto is fixed, friendships never change.
Heuristic baselines	The Oracle and the plan search are strong, but not proven optimal.
Sampled knowledge	Logic verdicts come from a finite sample of smaller worlds.
No human data	Every result comes from computer players; the playtest was dropped.

TEAM DUMBOCRACY

Individual contributions

G. Madhava Asrith Murthy AI23BTECH11007	Built the smart player (surveys, manifesto, planning) and ran the strategy experiments.
B. U. Deekshith Patel AI23BTECH11003	Built the electorate: regions, friendship network, preferences, zealots, drift and voting.
Vuppula Hemanth Reddy AI23BTECH11033	Designed what the player sees and buys: polls, surveys, workers, media and their prices.
Pakki Sai Kaushik AI23BTECH11017	Built the epistemic and temporal logic, the game engine and the GPU experiment harness.
Vishank Singh AI23BTECH11032	Built the web game: its screens, visualisations, debrief timeline and web API.

TEAM DUMBOCRACY

Thank you.

Questions?

github.com/saikaushhikp/CLOUT

Pakki Sai Kaushik · G. Madhava Asrith Murthy · Vuppula Hemanth Reddy · Vishank Singh · B. U. Deekshith Patel